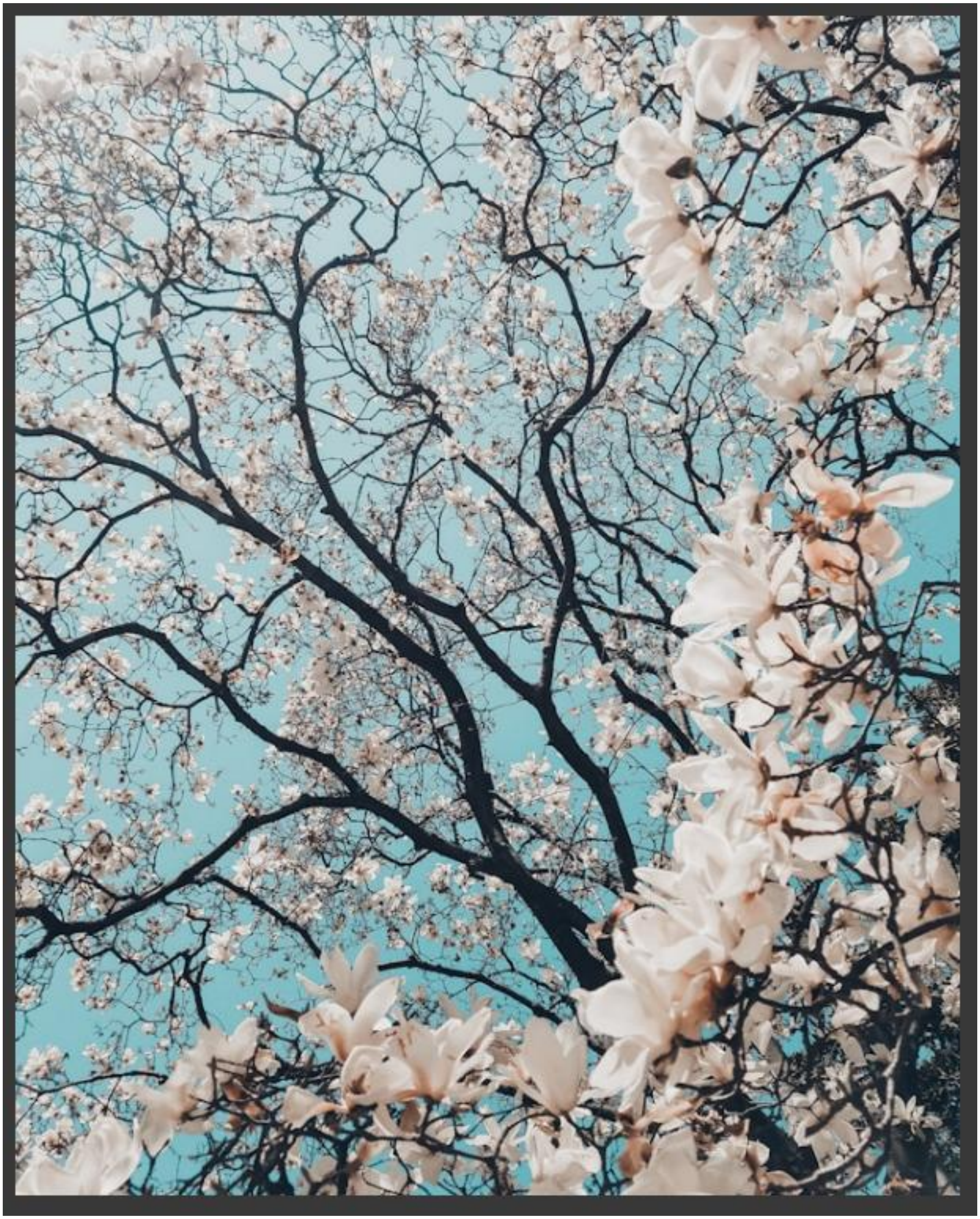
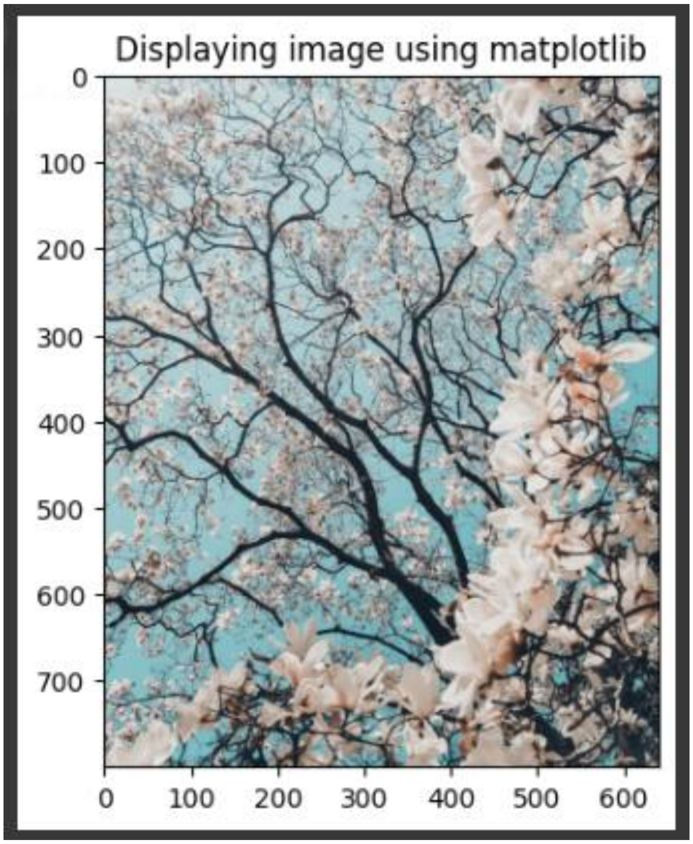
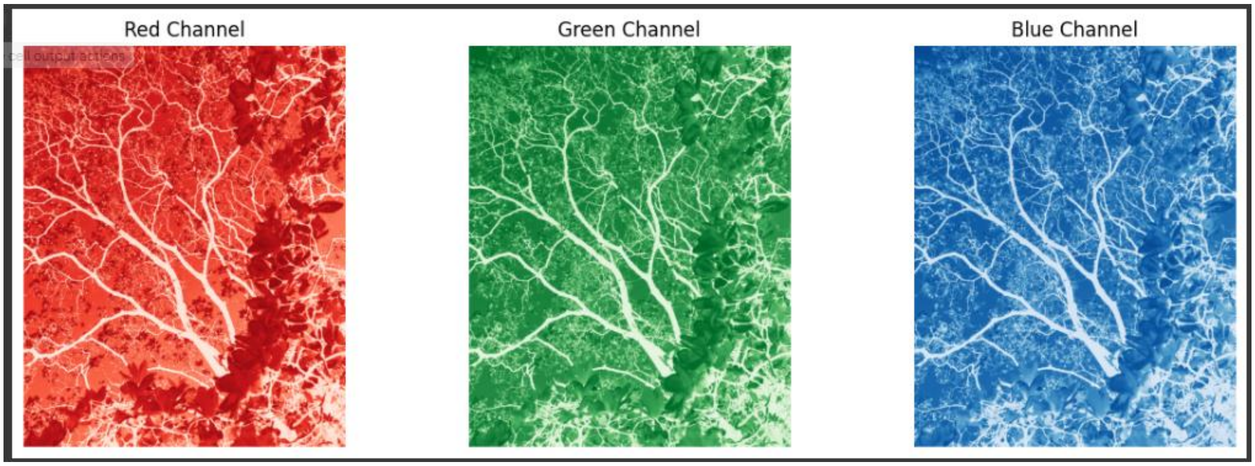
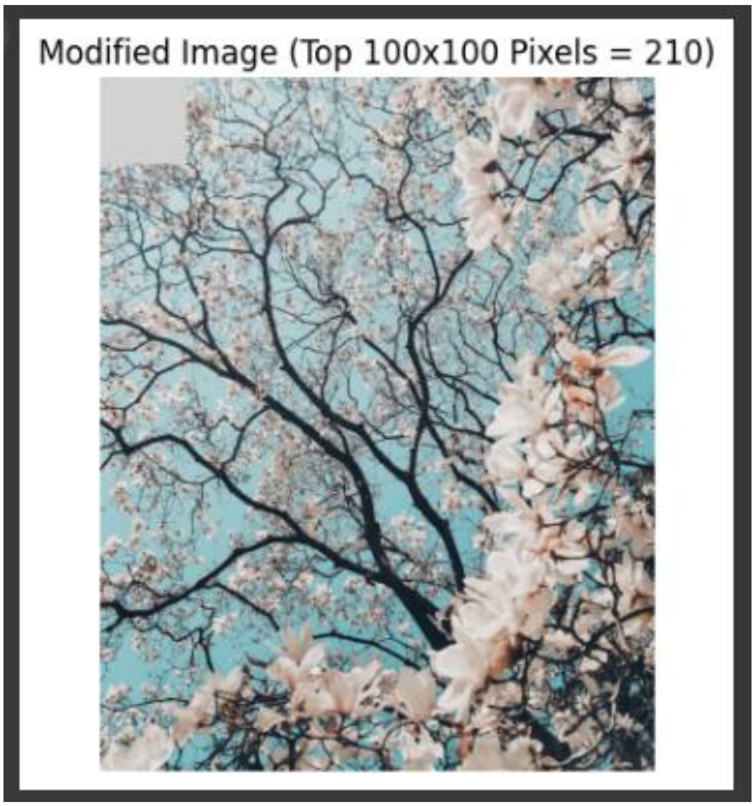
Exercise-1




Displaying image using matplotlib
0
100
200
300
400
500
600
700
0
100
200
300
400
500
600

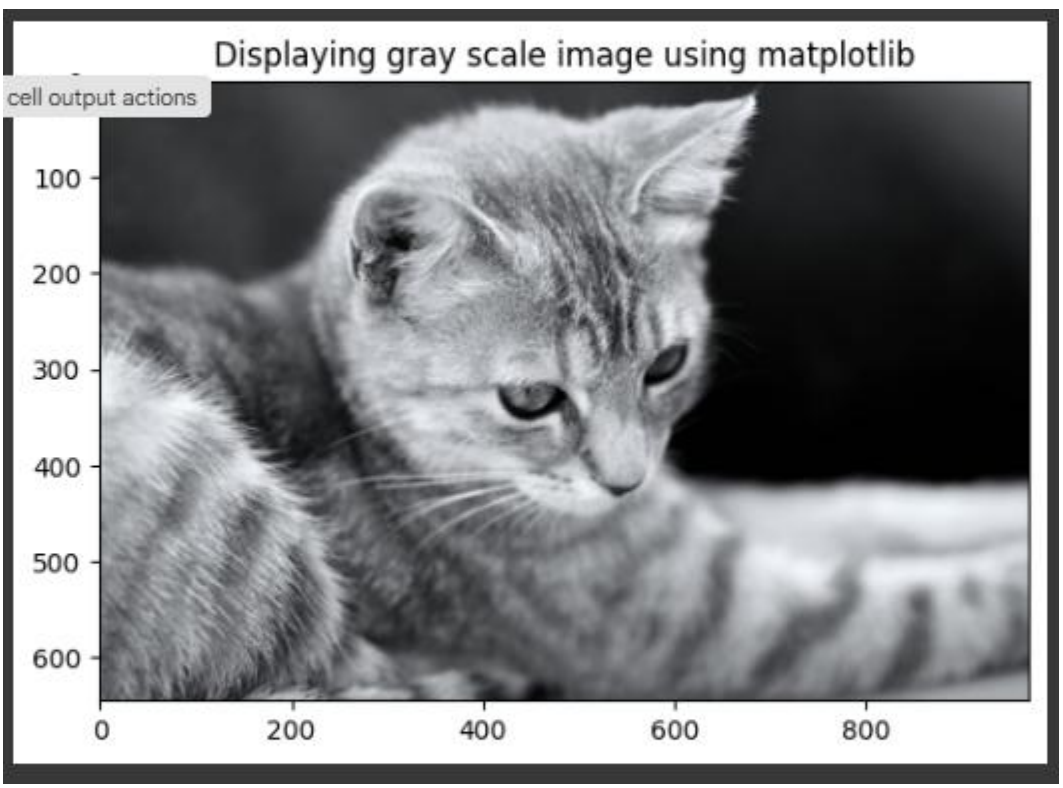

cell output actions

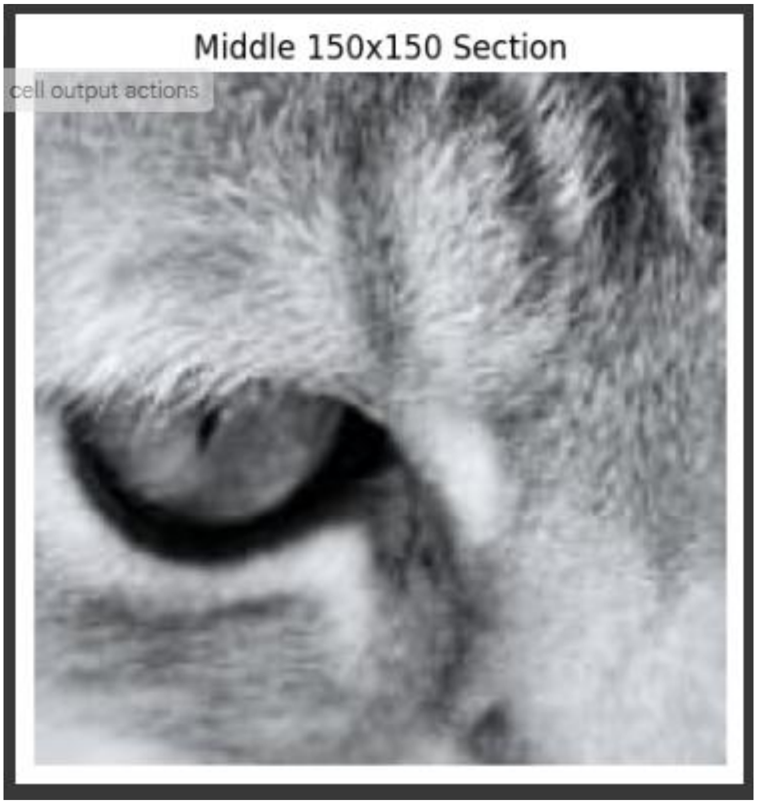

Red Channel
Green Channel
Blue Channel

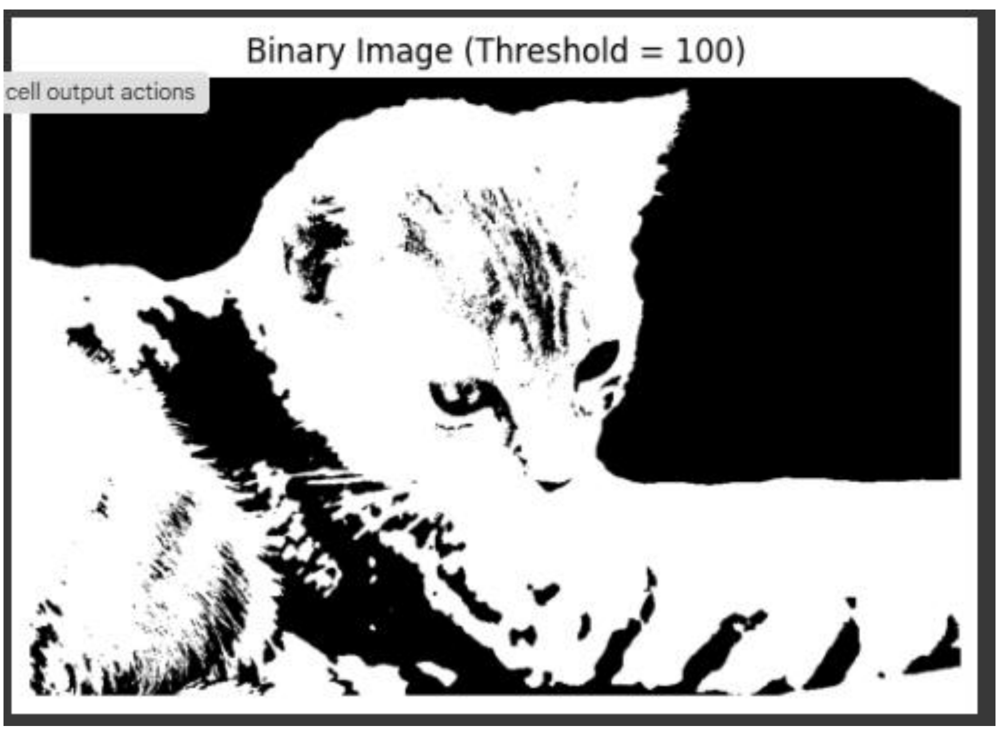

Modified Image (Top 100x100 Pixels = 210)

Exercise-2


cell output actions


Displaying gray scale image using matplotlib
cell output actions
100
200
300
400
500
600
0
200
400
600
800


Middle 150x150 Section
cell output actions


Binary Image (Threshold = 100)
cell output actions

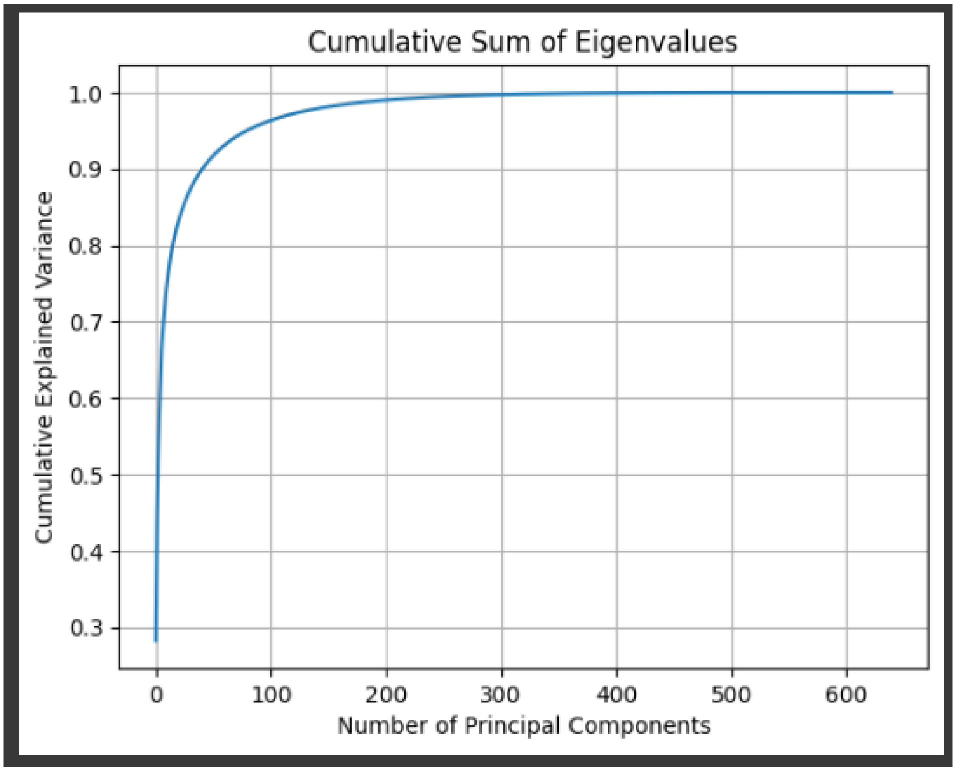
Rotated 90 Degrees Clockwise (NumPy)

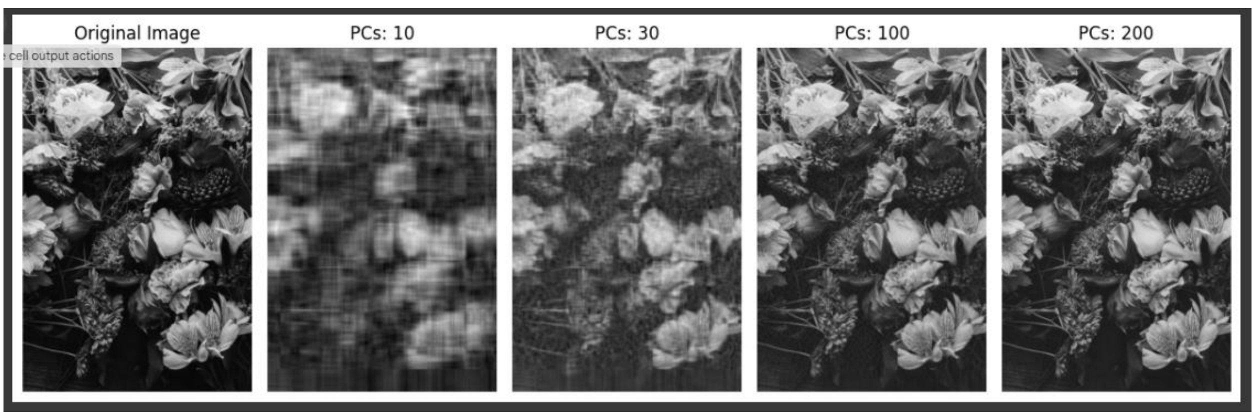
Grayscale to RGB Image

Exercise-3

```
Standardized Image Shape: (960, 640)
```


```
Covariance Matrix: [[3445.3867918  3362.80829857 3246.14817518 ...  563.02263643
   633.95263295  669.7745829 ]
 [3362.80829857 3572.21728254 3512.13596194 ...  627.29477863
   706.3168763   735.33176812]
 [3246.14817518 3512.13596194 3705.89348279 ...  717.00918926
   789.13192127  826.82782847]
 ...
 [ 563.02263643  627.29477863  717.00918926 ... 2156.46712613
  2065.81701316 1904.51223605]
 [ 633.95263295  706.3168763   789.13192127 ... 2065.81701316
  2238.47319148 2161.16346129]
 [ 669.7745829   735.33176812  826.82782847 ... 1904.51223605
  2161.16346129 2327.27246155]]
```





Evaluation

Principal Components - Observation

10 - Very blurry, major details lost

30 - Some details retained but still low quality

100 - Most details are clear

200 - Almost identical to the original